Instrumentation

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

1. A telescope is a light bucket and an image maker
2. Resolution: diffraction, seeing and pixels
3. Detectors count photons
4. Mounts, tracking and field rotation
5. Calibration turns counts into measurements
6. Choose the instrument for the question
7. Lecture + Problems: predict, calculate, check
8. Further practice and provenance

1. A telescope is a light bucket and an image maker

The two most important telescope numbers are aperture D and focal length f. Aperture determines how much light is collected and, in principle, how fine the diffraction limit is. Focal length determines image scale. The focal ratio is

$$N = \frac{f}{D}. \tag{1}$$

A 200 mm telescope with a 1000 mm focal length is f/5. For visual use, magnification is $M = f_{\text{objective}}/f_{\text{eyepiece}}$, and the exit pupil is D/M. An exit pupil larger than the dark-adapted eye pupil wastes light; a very small exit pupil makes the image dim and diffraction effects more noticeable.

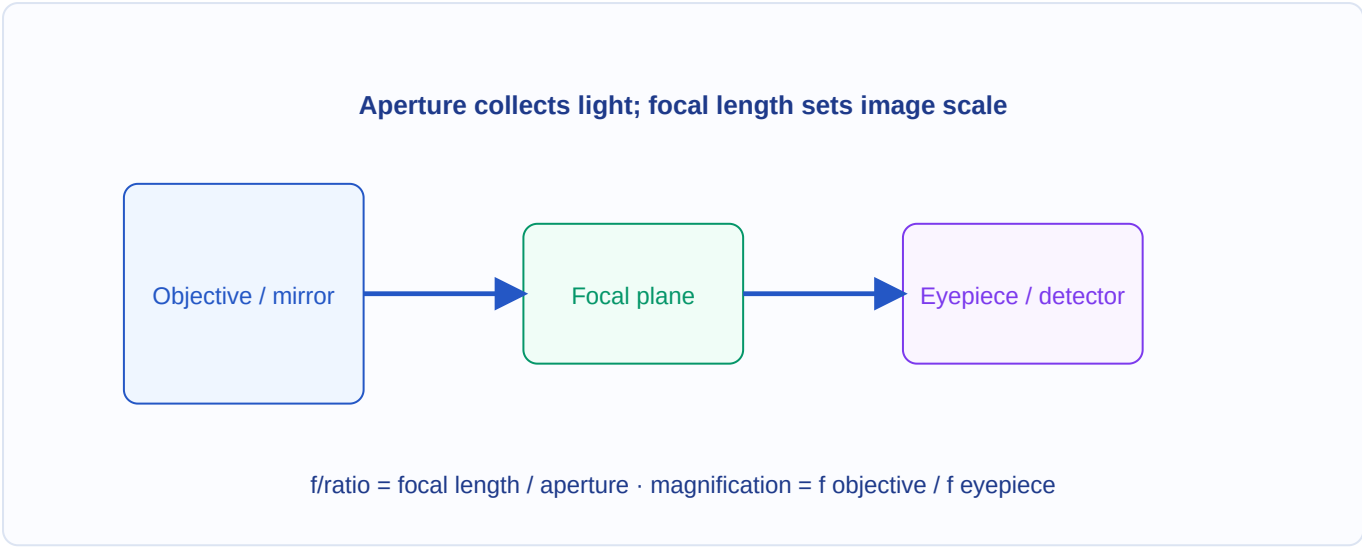


Figure 1. Aperture controls light grasp and diffraction; focal length and detector or eyepiece control scale and field.

2. Resolution: diffraction, seeing and pixels

For a circular aperture, the Rayleigh criterion is

$$\theta = 1.22 \frac{\lambda}{D}. \tag{2}$$

At 550 nm this is approximately $138.4 / D_{\mathrm{mm}}$ arcseconds. A 200 mm telescope has a diffraction limit near 0.69″. From the ground, atmospheric seeing often broadens the image to 1–3″, so the effective resolution is usually the worse of diffraction and seeing.

Detector pixel scale also matters:

$$s = 206.265 \, \frac{p_{\mu\mathrm{m}}}{f_{\mathrm{mm}}} \quad \mathrm{arcsec/pixel}. \tag{3}$$

Undersampling loses detail; oversampling spreads light over many pixels and can reduce signal-to-noise per pixel.

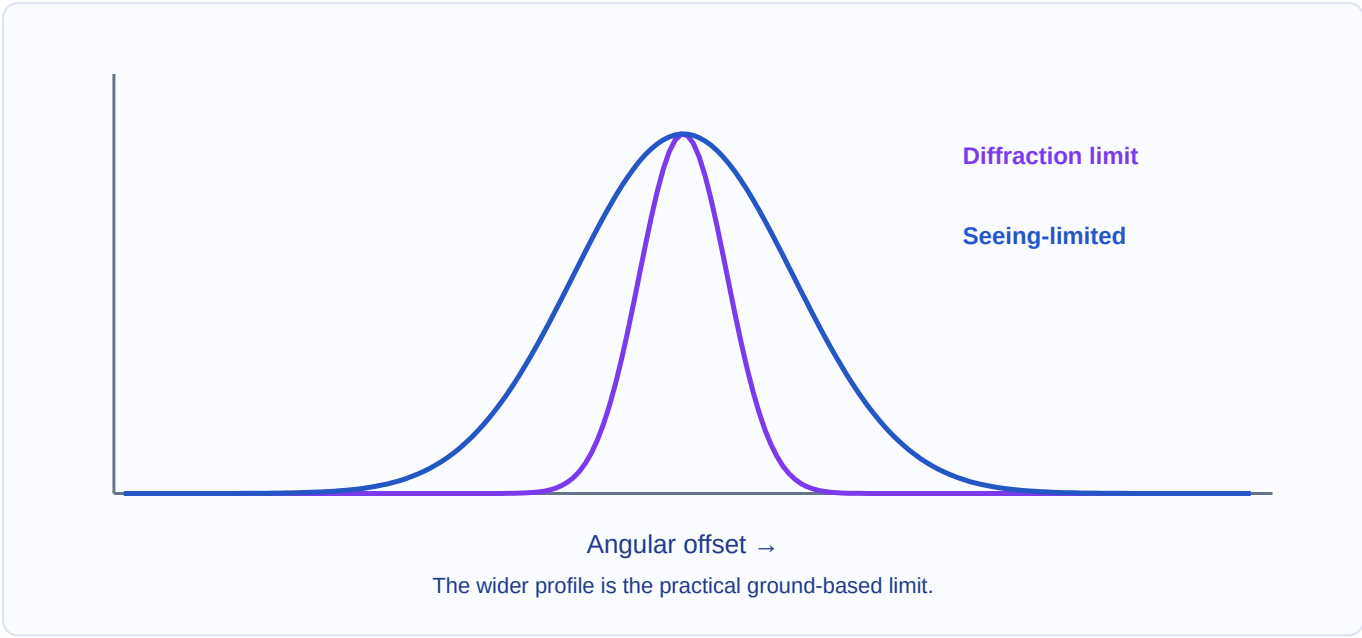


Figure 2. Adaptive optics, space telescopes or interferometry can recover finer angular resolution.

3. Detectors count photons

Modern detectors convert photons into electrons with quantum efficiency QE. The signal from a source is proportional to collecting area, exposure time, throughput and QE. Noise comes from the source itself, sky background, dark current and read noise. Here S, B, D are total electrons in the photometric aperture, $n_{ m pix}$ is the number of pixels in that aperture, and $\sigma_{ m read}$ is the read noise per pixel. For independent contributions,

$$\mathrm{SNR} = \frac{S}{\sqrt{S + B + D + n_{\mathrm{pix}}\sigma_{\mathrm{read}}^2}}, \tag{4}$$

In the source-dominated limit, SNR grows as $\sqrt{t}$. In the background-dominated limit, it still grows as $\sqrt{t}$, but with a smaller coefficient. Read noise is especially important for short exposures.

4. Mounts, tracking and field rotation

The sky appears to rotate at about 15″ per second of time near the celestial equator. An untracked telescope trails stars quickly. Equatorial mounts track by rotating about one axis aligned with Earth’s pole; alt-azimuth mounts need two-axis motion and may suffer field rotation. Guiding corrections keep the target on the same pixels during long exposures.

Tracking quality is part of the instrument. A large telescope with poor tracking can produce worse data than a smaller telescope with stable guiding.

5. Calibration turns counts into measurements

Raw frames contain instrumental signatures. Bias frames remove electronic offset; dark frames remove thermal signal; flat fields correct pixel-to-pixel sensitivity, vignetting and dust shadows. Standard stars or catalogue photometry set the zero point. Spectroscopy needs wavelength calibration and often flux calibration.

A measurement is not complete when the exposure ends. It is complete when the calibration chain and uncertainty budget are known.

6. Choose the instrument for the question

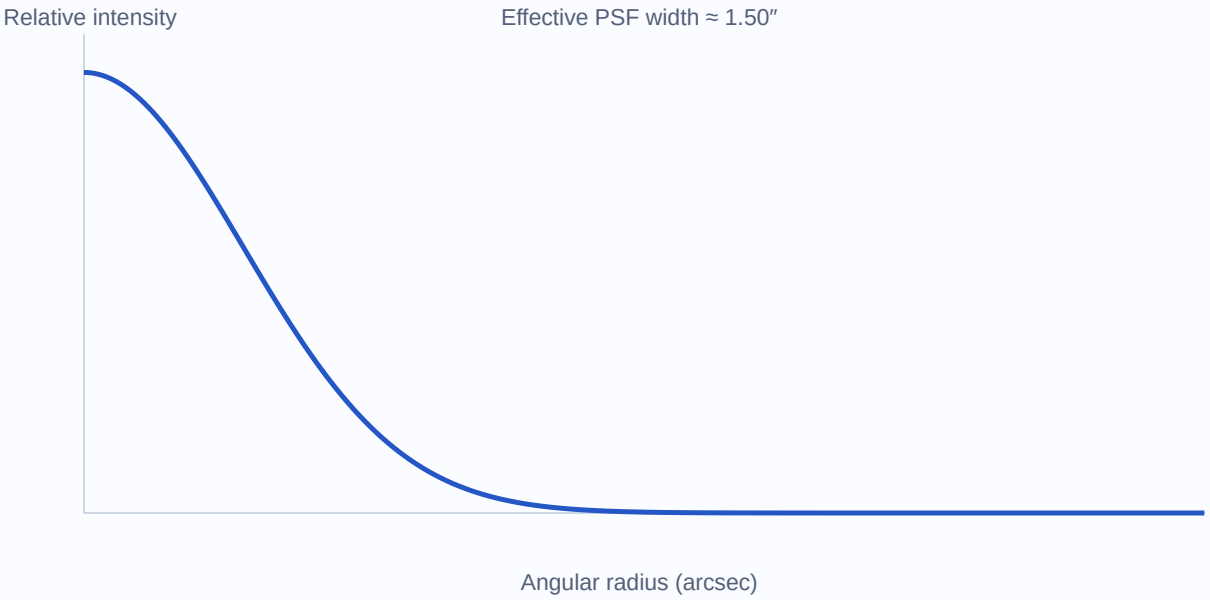
Visual observation, imaging, photometry and spectroscopy have different requirements. A double-star observer needs resolution and stable seeing; a variable-star observer needs stable photometry and time coverage; a spectroscopist needs throughput and wavelength stability. There is no universally “best” telescope.

INTERACTIVE EXPERIMENT

Aperture, focal length and the atmosphere.

Diffraction sets a theoretical limit; seeing usually dominates from the ground. Magnification does not create resolution that the aperture and atmosphere did not provide.

Focal ratio **f/5.0**	Magnification **50.0×**	Exit pupil **4.00 mm**
Rayleigh limit **0.69″**	Effective resolution **1.50″**	Approx. limiting mag **13.5**



Magnification = focal length / eyepiece focal length. Exit pupil = aperture / magnification. The atmosphere usually sets the practical resolution.

Try this. Double the aperture. What happens to the diffraction limit and light grasp? Then make the exit pupil larger than a dark-adapted eye pupil.

Snapshot of the experiment. Open the HTML edition to change the controls.

A reliable method. Write down the question first. Then choose aperture, focal ratio, detector, filter and mount. Check exit pupil, pixel scale, tracking and calibration before observing.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Basic telescope numbers. A telescope has $D = 200$ mm and $f = 1000$ mm. Find its focal ratio, magnification with a 20 mm eyepiece, and exit pupil.

Worked solution →

[1] Problem 2 — Diffraction and seeing. Estimate the Rayleigh limit of a 200 mm telescope at 550 nm. If the seeing is 1.5″, what effective resolution should you expect?

Worked solution →

[1] Problem 3 — Light grasp. How much more light does a 200 mm telescope collect than a 50 mm telescope? What is the corresponding gain in limiting magnitude, ignoring other losses?

Worked solution →

[2] Problem 4 — Pixel scale. A camera has 5 μm pixels on a 1000 mm focal-length telescope. Find the pixel scale. Is a 1.5″ seeing disk well sampled?

Worked solution →

[2] Problem 5 — Signal-to-noise. An exposure yields 10,000 source electrons, 2500 sky electrons and 100 dark electrons in the aperture. The total read-noise variance in that aperture is 25 electrons² (equivalently, one pixel with 5-electron read noise). Estimate the SNR.

Worked solution →

[1] Problem 6 — Doubling the exposure. In the source-dominated limit, by what factor does SNR increase when exposure time is doubled? Does doubling the telescope aperture have the same effect on SNR for a point source in the same exposure time?

Worked solution →

[2] Problem 7 — Trailing without tracking. If the sky moves at 15″/s, how long a trail is produced by a 60 s untracked exposure near the celestial equator? Why is this usually unacceptable?

Worked solution →

[2] Problem 8 — Photon budget. A zero-magnitude source delivers 10^{10} photons $m^{-2}\ s^{-1}\ \mu m^{-1}$. A 200 mm telescope observes a 10th-magnitude star through a 0.1 μm filter with QE 0.8 for 10 s. Estimate the detected electrons.

Worked solution →

[1] Problem 9 — Calibration frames. State the purpose of bias, dark and flat frames. Which correction addresses pixel-to-pixel sensitivity and dust shadows?

Worked solution →

[3] Problem 10 — Choose an instrument. You want to measure a 1.0″ double star from a site with 1.0″ seeing. Would a 150 mm telescope reach the diffraction requirement? Would it guarantee a clean split? What would help more than magnification alone?

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.